```
=== SECTION 1: DATA INPUT ===

*** SYNTHETIC DEMONSTRATION MODE ***
These signals are NOT real motor data.
Set useRealData=true after downloading the CWRU dataset.

Signal info:

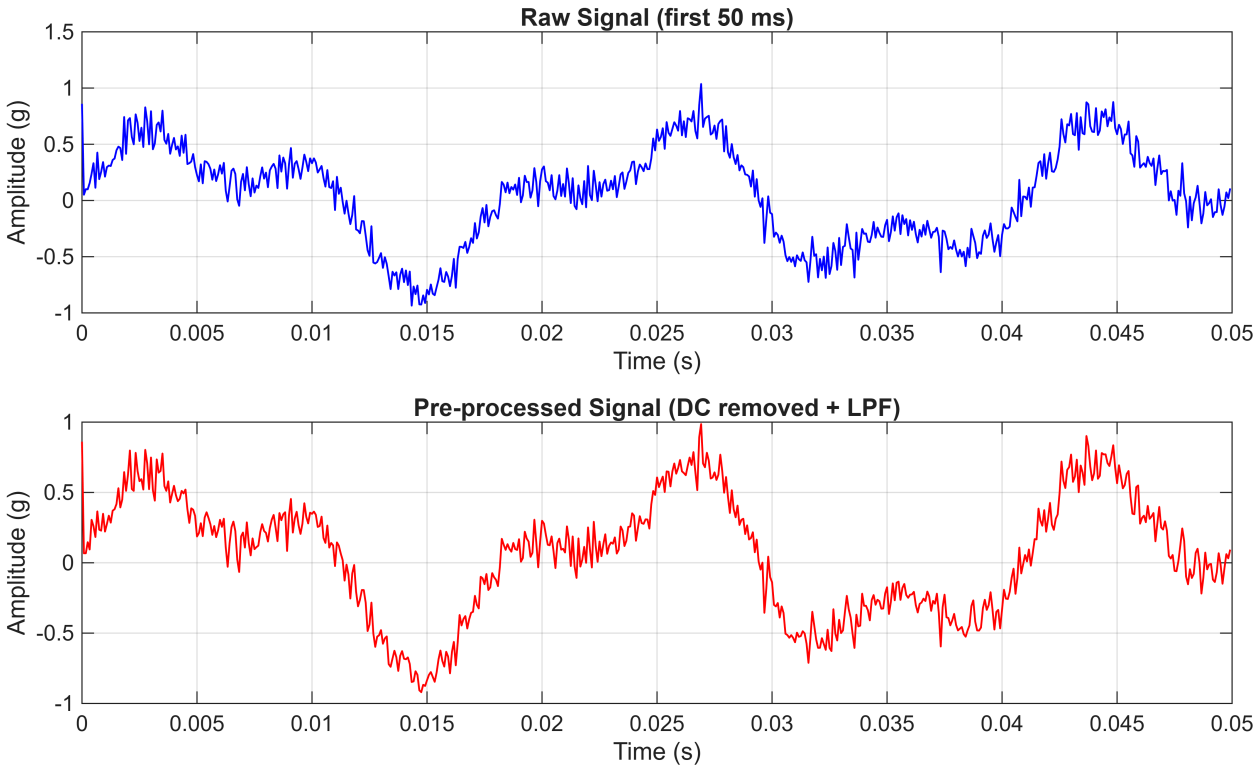
  Sampling frequency : 12000 Hz

  Signal duration    : 2.000 seconds

  Total samples      : 24001

=== SECTION 2: PRE-PROCESSING ===

DC removed and Butterworth LPF applied (fc = 5000 Hz, order 4).
```




```
Saved: fig1_raw_vs_preprocessed.png

=== SECTION 3: TIME-DOMAIN FEATURES ===

Feature              Normal          Fault

--------------------------------------------------

Mean                 -0.00000        0.00000

RMS                  0.36360         0.42270

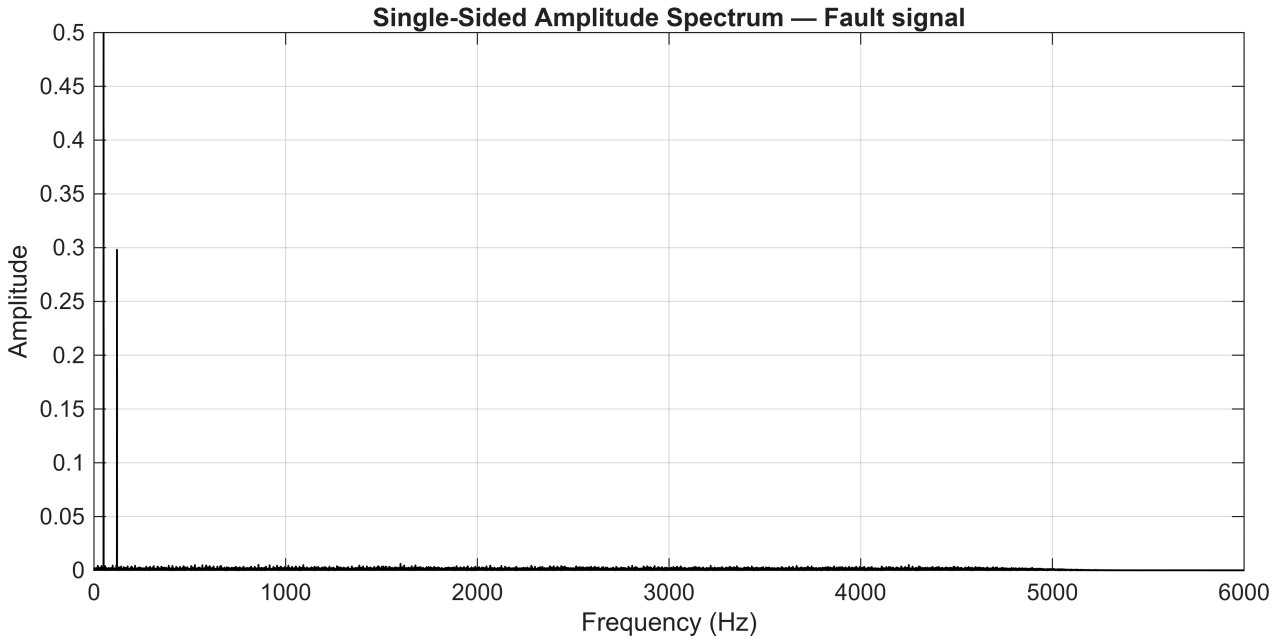
Std Dev              0.36361         0.42271

Peak                 0.69179         1.55735

Crest Factor         1.90261         3.68429

Kurtosis             1.64997         2.19483

=== SECTION 4: FREQUENCY-DOMAIN ANALYSIS ===
```




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Saved: fig2_fft_spectrum.png

Top dominant frequencies:

  Peak 1: 50.0 Hz  (amplitude = 0.4994)
  Peak 2: 120.0 Hz  (amplitude = 0.2982)

Fault-related spikes at non-harmonic frequencies suggest bearing defects.

=== SECTION 5: FEATURE EXTRACTION ===

Feature vector (active signal):

  RMS          : 0.4227
  StdDev       : 0.4227
  Peak         : 1.5573
  CrestFact    : 3.6843
  Kurtosis     : 2.1948
  DomFreq1     : 49.9979
  DomFreq2     : 119.9950
  DomFreq3     : 0.0000

=== SECTION 6: CONDITION ASSESSMENT ===

Rule-based thresholds:

  RMS threshold    : 0.150 g

  Crest threshold  : 4.50

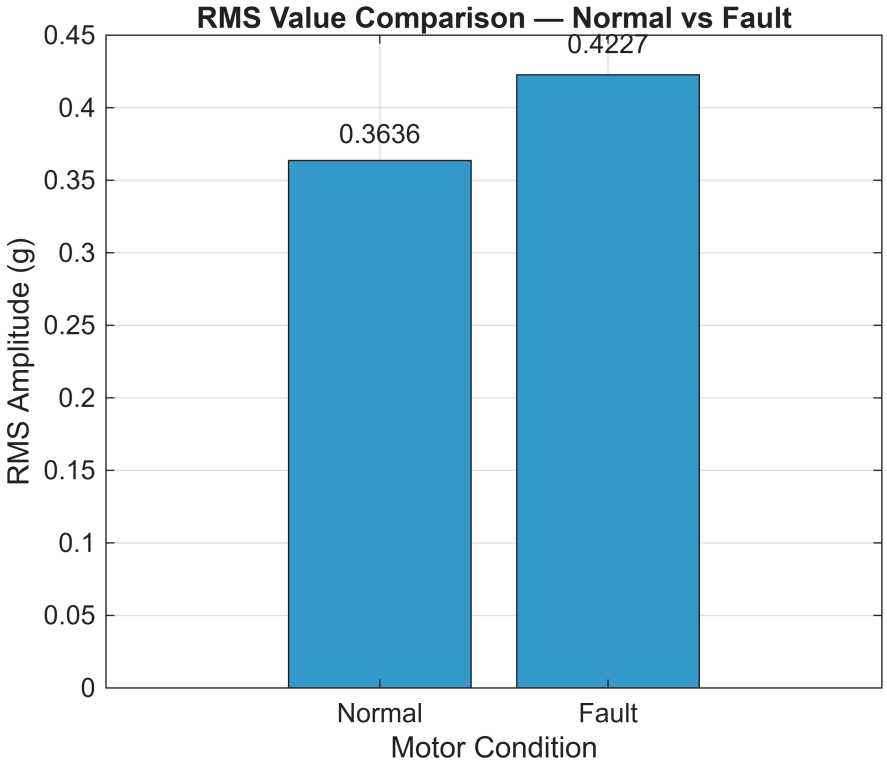
  Kurtosis threshold: 4.00

Flags raised    : High RMS

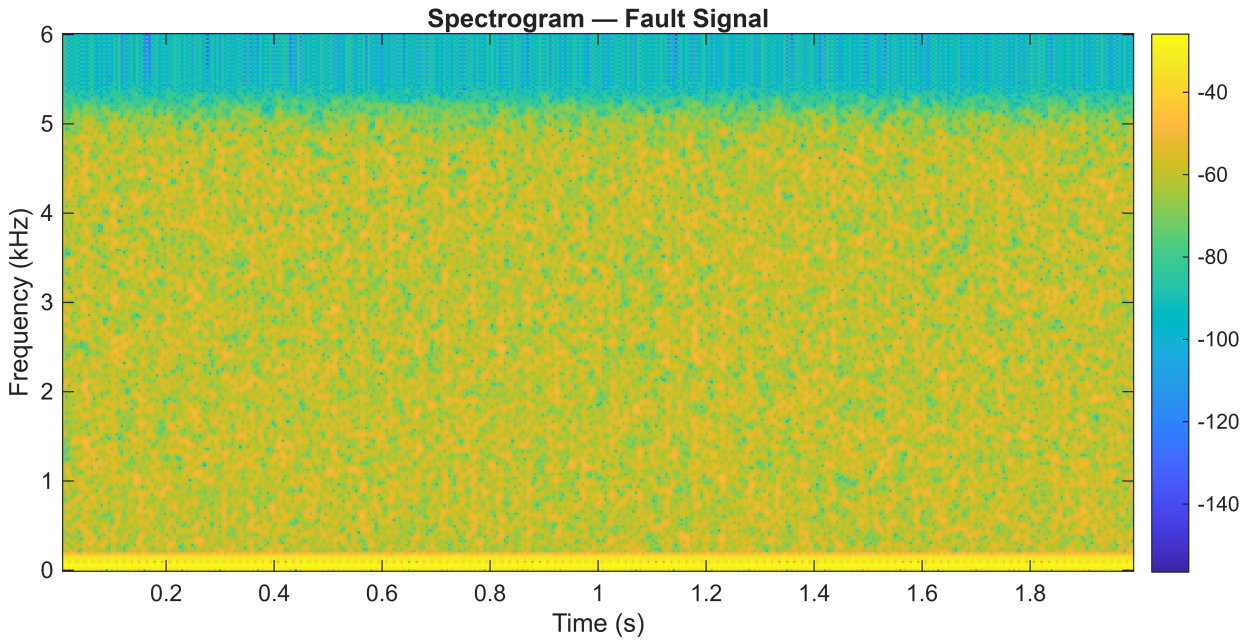
Rule-based verdict: [ Requires further inspection ]

Note: This is an algorithmic estimate, not a guaranteed mechanical diagnosis.

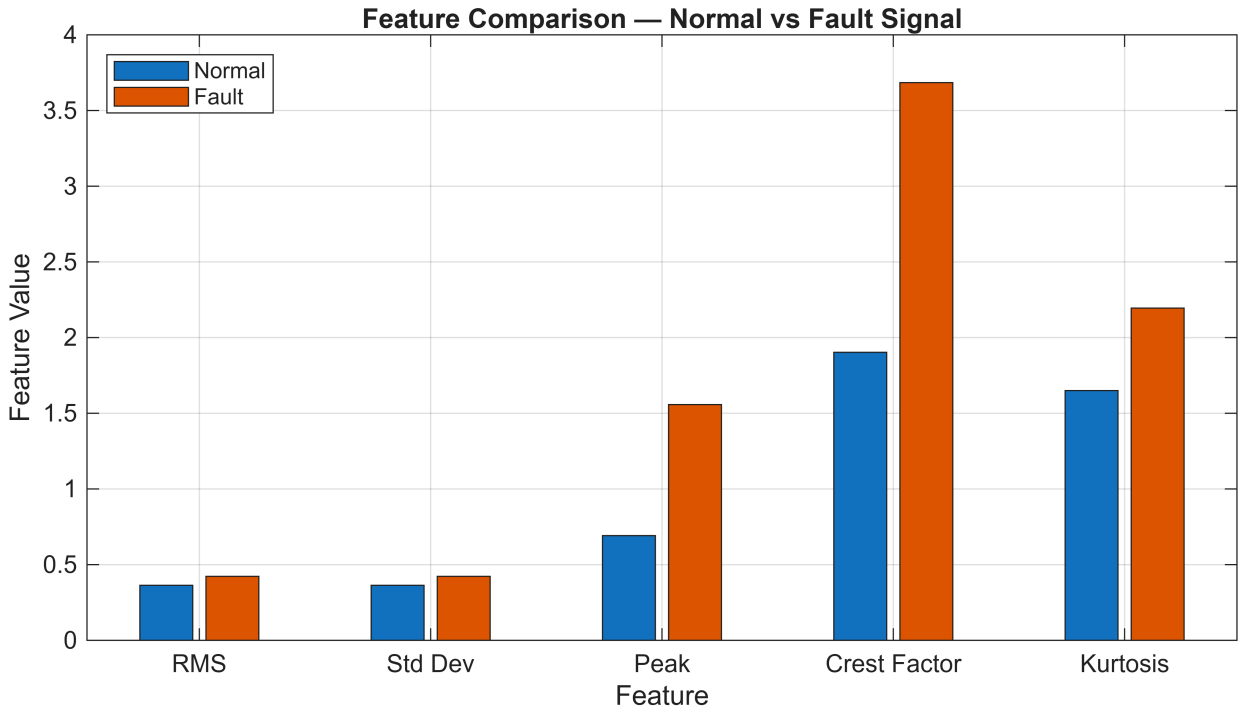
=== SECTION 7: GRAPH GENERATION ===
```




```
Saved: fig3_rms_comparison.png
```




```
Saved: fig4_spectrogram.png
```




```
Saved: fig5_feature_comparison.png

=== SECTION 8: PERFORMANCE EVALUATION ===

SVM evaluation skipped (useSVM=false or insufficient data).
To evaluate: set useSVM=true, ensure CWRU files are available.
Performance formula reference:
  Accuracy  = (TP+TN)/(TP+TN+FP+FN)
  Precision = TP/(TP+FP)
  Recall    = TP/(TP+FN)
  F1        = 2*Prec*Rec/(Prec+Rec)
  FAR       = FP/(FP+TN)

=== ALL SECTIONS COMPLETE ===

Generated figures:

  fig1_raw_vs_preprocessed.png

  fig2_fft_spectrum.png

  fig3_rms_comparison.png

  fig4_spectrogram.png

  fig5_feature_comparison.png

Saved result to: analysis_result.mat

--- FINAL VERDICT ---

Condition assessed as: [ Requires further inspection ]

This is an algorithmic estimate. Not a guaranteed mechanical diagnosis.
```